# Kerning & Typography Stress Test

This fixture validates that glyph clustering and word reconstruction handle extreme tracking, mixed font sizes on identical baselines, and varied weights without collapsing words into spaced-out characters.

## Font Family & Weight Variations

Regular weight body text in the default typeface.

**Bold weight body text for emphasis and headings.**

Italic weight for citations and foreign phrases.

Bold-italic weight for maximum emphasis.

### Mixed Font Sizes on Shared Baseline

Text with small 8pt and large 14pt and normal 10pt sharing one baseline tests y-coordinate grouping.

Another mixed line: tiny HUGE normal all on the same line.

## Monospace Code Blocks

The following code uses a monospace face and must keep indentation intact:

```
def fibonacci(n):
if n <= 1:
return n
return fibonacci(n-1) + fibonacci(n-2)
result = fibonacci(10) # Should be 55
print(f"fib(10) = {result}")
```

Code above must not be rewritten; it contains intentional formatting tokens.

## Natural Word Spacing

The quick brown fox jumps over the lazy dog. This sentence tests natural word spacing: gaps between words should be spaces, not character splits.

Pack my box with five dozen liquor jugs. Another pangram for spacing validation.

Typography ligatures: fi fl ffi ffl should remain as combined glyphs or separate characters without extra spacing artifacts.

### Colored Typography

Color test: red text α → Ω and blue text ∑ ≠ ± must keep RGB values. Colors are structural, not content.

# Low-Level Kerning Validation

The following lines are drawn with precise glyph positioning to test kerning extraction. Each line's words must reconstruct correctly.

The quick brown fox jumps over the lazy dog

Kerning test: AVATAR WAVE To

Typography: fi fl ff ffi ffl ligatures

Mixed sizes on same baseline

Natural word spacing must survive

## End of Kerning Fixture

If any exported text contains spaced-out characters like "t h i s", the extractor's glyph clustering is broken.

Normal spacing: Hello World Test
Tight tracking: Hello World Test
Wide tracking: Hello World Test
Small LARGE normal all same baseline
AVATAR WAVE To Va We kerning pairs
The end of manual kerning tests.

Low-Level Text Operators
Kerning TJ array keeps words whole
Tracking with Tc must not split words
Negative tracking keeps words whole
Per glyph Tm positioning still forms words
Wide word spacing keeps letters intact
Word gap from geometry alone